GEOMETRY QUEST · WORLD 1

Points & Lines

WORKSHEET

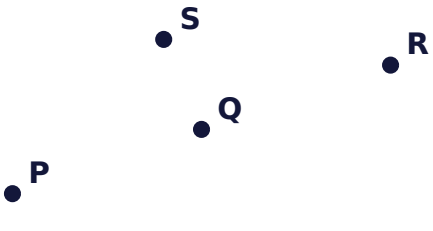
Name _____ Date _____ Stars I give myself: ☆ ☆ ☆

1 Name each one: line segment, ray or line.



_____ _____ _____

2 Three of these points are collinear (on the same straight line). Which three? Use a ruler to check.



_____

3 Measure line segment AB with a ruler. How long is it?



_____ cm

4 How many different straight lines can you draw through two points?

_____

5 Draw a line segment exactly 6 cm long. Label its ends C and D.

6 Real-life hunt: name something at home that is like a ray, and something like a line segment.

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0

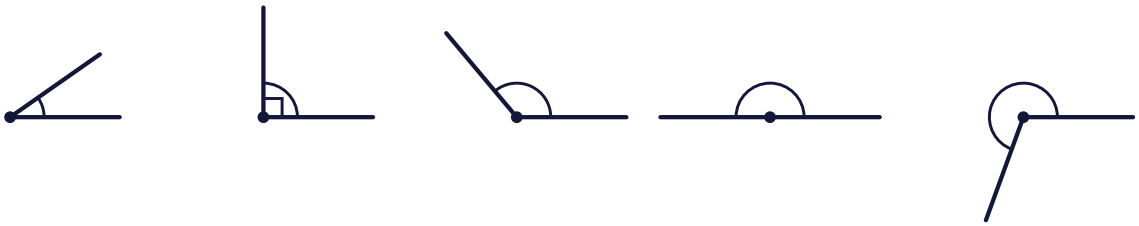
GEOMETRY QUEST · WORLD 2

Angles

WORKSHEET

Name _____ Date _____ Stars I give myself: ☆ ☆ ☆

1 Write acute, right, obtuse, straight or reflex under each angle.



_____ _____ _____ _____ _____

2 What angle do the hands of a clock make at each time? (Think: 360° ÷ 12 = 30° per hour.)

3:00 → _____ ° **6:00 →** _____ ° **2:00 →** _____ ° **4:00 →** _____ °

3 Fill in the gaps.

Two angles that add up to 90° are called _____

Two angles that add up to 180° are called _____

The complement of 35° is _____

The supplement of 110° is _____

4 Turning. You face North. Which way are you facing after each turn (from North each time)?

a quarter turn clockwise (90°) _____

a half turn (180°) _____

270° clockwise _____

5 Skater challenge: a “540” is how many full turns?

_____

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0

GEOMETRY QUEST · WORLD 3

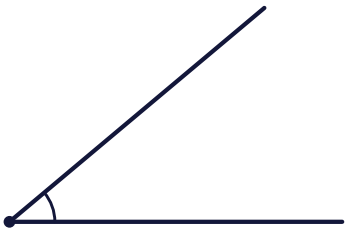
Protractor Skills

WORKSHEET

Name _____ Date _____ Stars I give myself: ☆ ☆ ☆

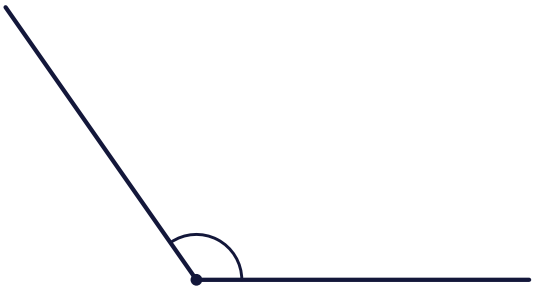
1 Measure each angle with a protractor. Put the centre on the corner and count up from the 0 on one arm. (Drawn to size: print at 100%.)

a)



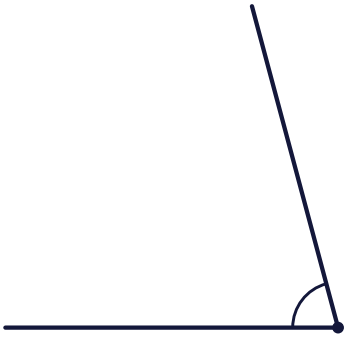
_____ °

b)



_____ °

c)



_____ °

d)



_____ °

2 Use your protractor to draw these angles on the lines below. Mark the angle with an arc.



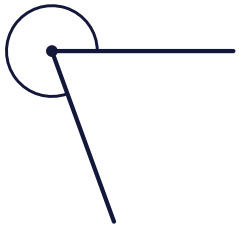
a) 50°



b) 135°

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0

3 This is a reflex angle. Measure it. (Tip: measure the small angle, then take it away from 360°.)



_____ °

4 Sam measured an angle and wrote 140°. But the angle is clearly smaller than a right angle. What went wrong, and what is the real size?

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0

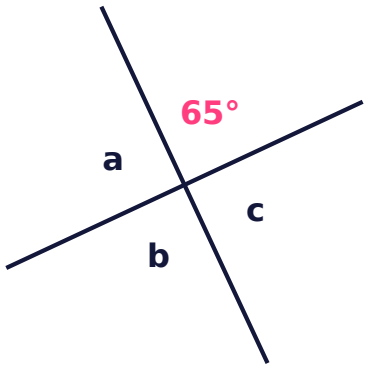
GEOMETRY QUEST · WORLD 4

Angle Rules

WORKSHEET

Name _____ Date _____ Stars I give myself: ☆ ☆ ☆

1 Two straight lines cross. Find a, b and c. Give a reason for each. (Not drawn to scale.)

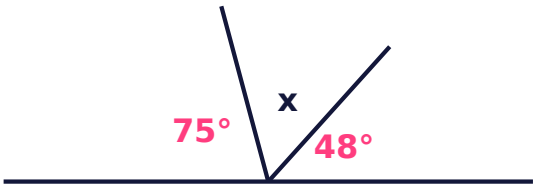


a = _____

b = _____

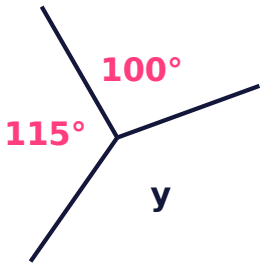
c = _____

2 These angles sit on a straight line. Find x.



x = _____ °

3 These angles meet at a point. Find y.



y = _____ °

4 Four equal angles meet at a point. How big is each one?

_____ °

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0

5 Open a pair of scissors. Explain why the angle between the blades always equals the angle between the handles.

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0

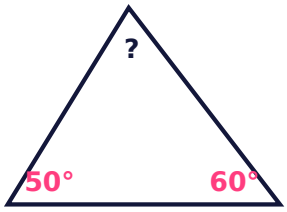
GEOMETRY QUEST · WORLD 5

Triangles

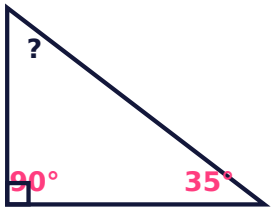
WORKSHEET

Name _____ Date _____ Stars I give myself: ☆ ☆ ☆

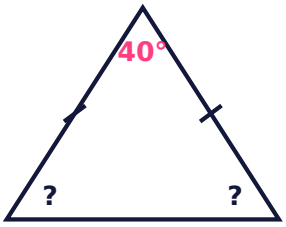
1 Find the missing angle in each triangle. (Not drawn to scale.)



_____



_____



_____

2 Classify each triangle by its sides (equilateral, isosceles, scalene) and its angles (acute, right, obtuse).

sides 5, 5, 5 cm _____

angles 90°, 45°, 45° _____

angles 120°, 40°, 20° _____

3 Area of a triangle = ½ × base × height. Work these out.

base 8 cm, height 5 cm	base 12 cm, height 7 cm	base 9 cm, height 4 cm
_____ cm²	_____ cm²	_____ cm²

4 An equilateral triangle has sides of 7 cm. What is its perimeter?

_____ cm

5 Experiment: draw any big triangle. Measure its three angles with a protractor and add them up. What do you get?

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0

GEOMETRY QUEST · WORLD 6

Polygons

WORKSHEET



Name _____ Date _____ Stars I give myself: ☆ ☆ ☆

1 Name the polygon with this many sides.

5 sides	**6 sides**	**8 sides**	**12 sides**
_____	_____	_____	_____

2 Angle total = (sides − 2) × 180°. Find the total of the inside angles.

hexagon _____ °

octagon _____ °

decagon _____ °

3 Each inside angle of a regular polygon = angle total ÷ number of sides. Find one angle of a regular pentagon and a regular hexagon.

pentagon _____ °

hexagon _____ °

4 True or false?

A square is a rectangle. _____

Every rectangle is a square. _____

A rhombus has four equal sides. _____

A trapezium has two pairs of parallel sides. _____

5 Find: (a) the area of a parallelogram with base 9 cm and height 4 cm; (b) the perimeter of a 7 cm by 3 cm rectangle.

(a) _____ cm²

(b) _____ cm

6 Polygon hunt: find three polygons at home or outside. Name each one and say where you found it.

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0

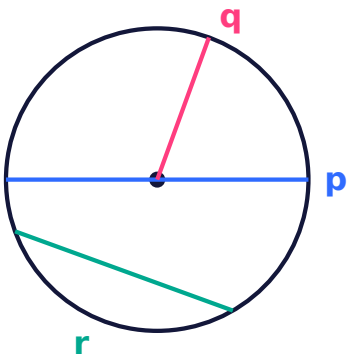
GEOMETRY QUEST · WORLD 7

Circles & π

WORKSHEET

Name _____ Date _____ Stars I give myself: ☆ ☆ ☆

1 Which line is the radius, which is the diameter, and which is a chord?



radius: _____

diameter: _____

chord: _____

2 Fill in: a radius of 4 cm means a diameter of ___ cm. A diameter of 15 cm means a radius of ___ cm.

_____ cm _____ cm

3 Circumference = π × diameter. Use π ≈ 3.14. Find the circumference when the diameter is 10 cm, and when it is 50 cm.

_____ cm _____ cm

4 Area = π × radius × radius. Find the area when the radius is 5 cm, and when it is 10 cm.

_____ cm² _____ cm²

5 A bike wheel is 70 cm across. How far does the bike travel in 10 full turns of the wheel? Give your answer in metres.

_____

6 Kitchen experiment: wrap string round a tin, measure it, then measure across the tin. Divide the first by the second.

Around: _____

Across: _____

Around ÷ across = _____

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0

GEOMETRY QUEST · WORLD 8

3D Shapes & Nets

WORKSHEET

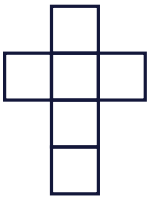
Name _____ Date _____ Stars I give myself: ☆ ☆ ☆

1 Complete the table. Then check that faces + vertices − edges = 2 for every shape.

shape	faces	edges	vertices	F + V − E
cube	6	12	8	
cuboid				
triangular prism				
square-based pyramid				
tetrahedron				

2 Which of these nets fold up into a cube? Tick them. (Imagine folding, or copy them onto squared paper and try.)

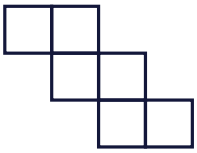
A



B



C



D



3 Name the 3D shape.

I have 6 square faces. _____

1 square face + 4 triangles meeting at a point. _____

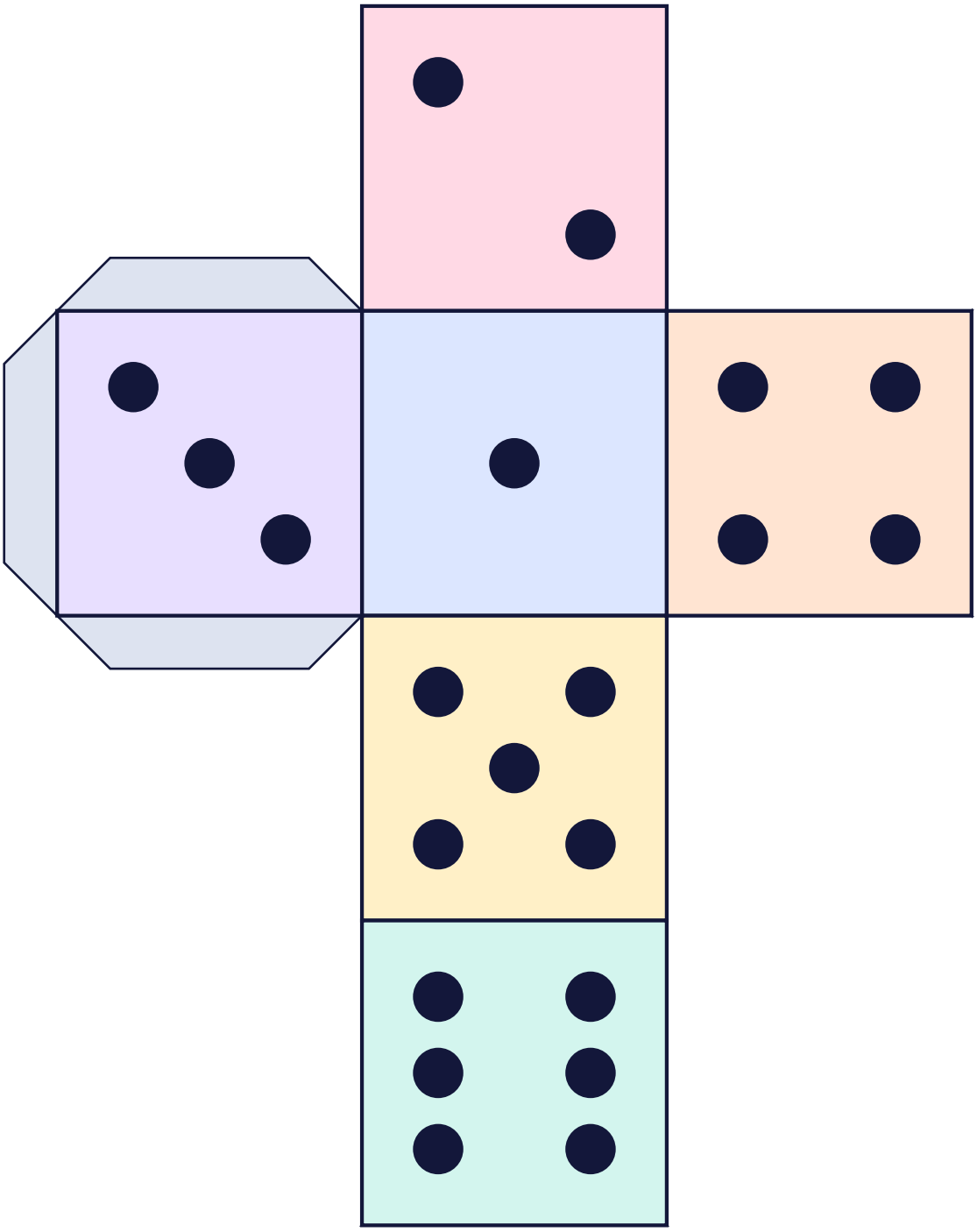
I have 2 flat circles and 1 curved surface. _____

I have one curved surface and no edges at all. _____

4 Cut out the dice net on the next page, fold along the lines and glue the tabs. Opposite faces of a real dice add up to 7. Do yours?

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0

Cut out, fold along every line, glue the grey tabs. Build your own dice!



Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0

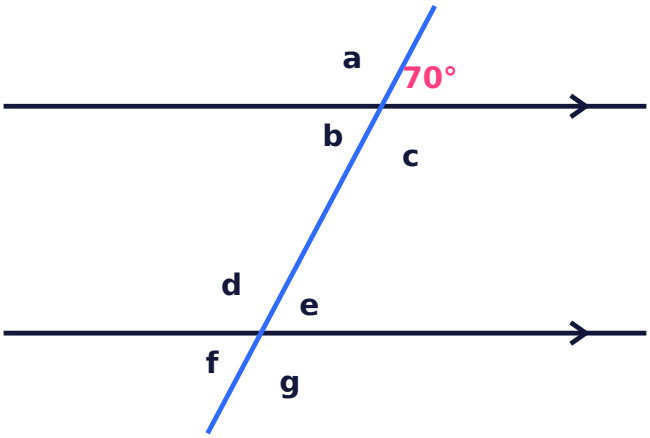
GEOMETRY QUEST · WORLD 9

Parallel Lines

WORKSHEET

Name _____ Date _____ Stars I give myself: ☆ ☆ ☆

1 The two rails are parallel. One angle is 70°. Find angles a to g. (Not drawn to scale.)



a = _____
b = _____
c = _____
d = _____
e = _____
f = _____
g = _____

2 Use the diagram. Which kind of angle pair is each one: corresponding (F), alternate (Z) or co-interior (C)?

70° and e _____
b and e _____
b and d _____

3 Two co-interior angles: one is 115°. What is the other?

_____ °

4 Parallel hunt: find four pairs of parallel lines at home (think shelves, stairs, window frames).

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0

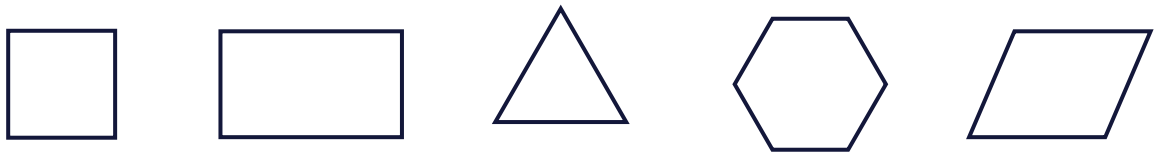
GEOMETRY QUEST · WORLD 10

Symmetry

WORKSHEET

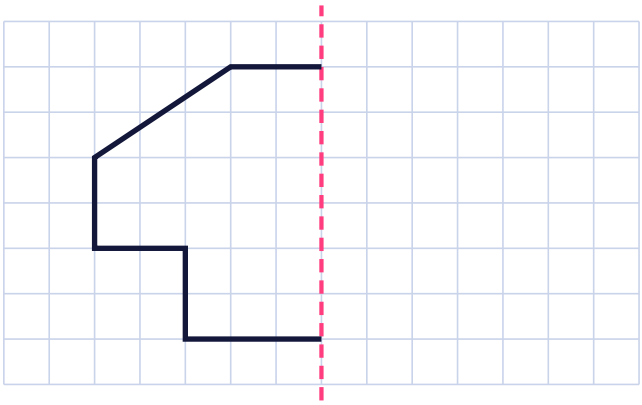
Name _____ Date _____ Stars I give myself: ☆ ☆ ☆

1 Draw all the lines of symmetry on each shape. How many does each have?



square	rectangle	equilateral △	reg. hexagon	parallelogram
_____	_____	_____	_____	_____

2 Complete the picture so the dashed line is a line of symmetry.



3 What is the order of rotational symmetry? (How many times does it fit onto itself in one full turn?)

letter S	letter H	equilateral △	square	recycling sign
_____	_____	_____	_____	_____

4 Circle the capital letters that have a vertical line of symmetry: A B F H J M P T U W Y Z

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0

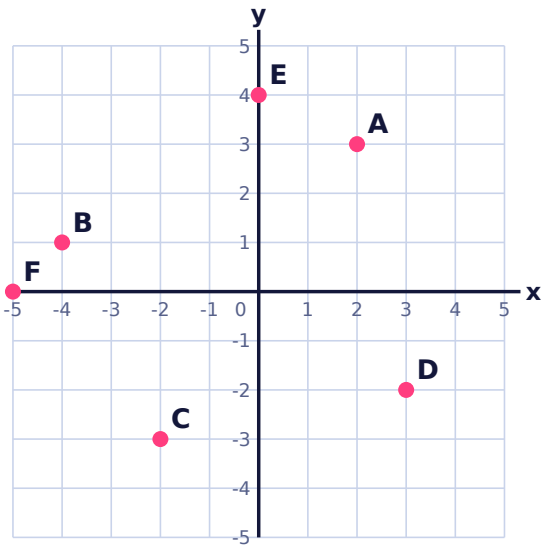
GEOMETRY QUEST · WORLD 11

Coordinates

WORKSHEET

Name _____ Date _____ Stars I give myself: ☆ ☆ ☆

1 Write the coordinates of each point. Remember: across first, then up (x, y).



A = (,)

B = (,)

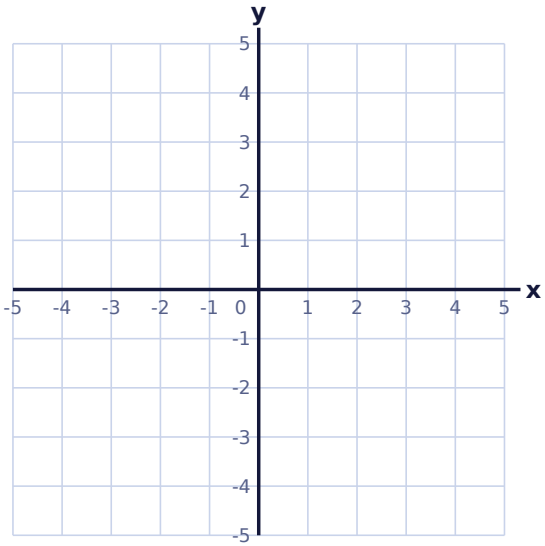
C = (,)

D = (,)

E = (,)

F = (,)

2 Plot these points and join them in order: (−3, −2), (3, −2), (3, 2), (−3, 2), back to the start. What shape is it?



Shape: _____

3 A square has corners at (1, 1), (4, 1) and (4, 4). Where is the fourth corner?

_____

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0

4 Which quarter (quadrant) of the grid is each point in: top-left, top-right, bottom-left or bottom-right?

(−2, 5) _____ **(4, −1)** _____ **(−3, −3)** _____

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0

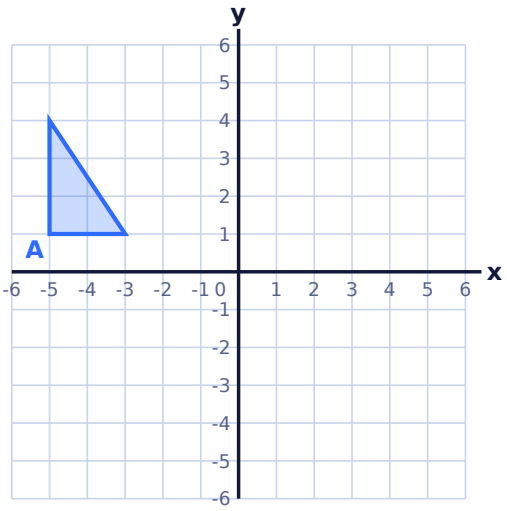
GEOMETRY QUEST · WORLD 12

Transformations

WORKSHEET

Name _____ Date _____ Stars I give myself: ☆ ☆ ☆

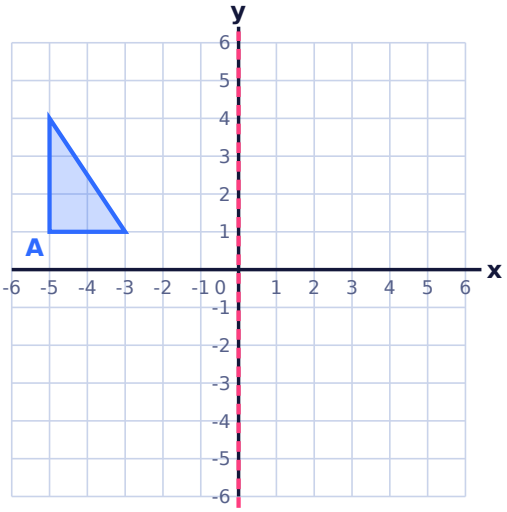
1 Translate the triangle 6 squares right and 4 squares down. Label the image A′.



Write the coordinates of A′:

_____

2 Reflect the triangle in the y-axis (the pink mirror line).

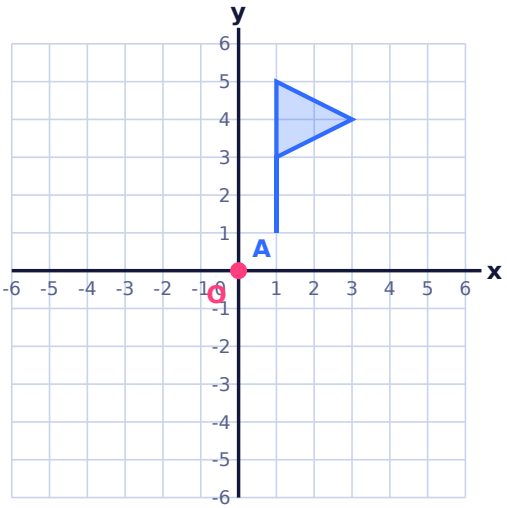


Write the coordinates of A′:

_____

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0

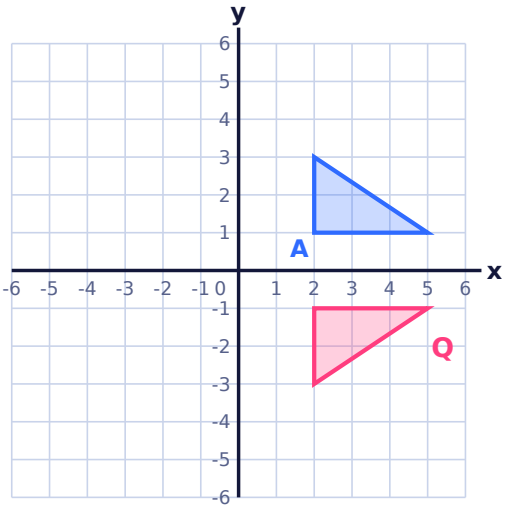
3 Rotate the flag 90° clockwise about the origin O. (Tracing paper helps!)



Write the coordinates of A′:

_____

4 Shape A was moved onto shape Q with one transformation. Describe it fully.



Describe the transformation:

_____

5 After a translation, reflection or rotation, what stays the same about the shape?
Tick: size ☐ shape ☐ angles ☐ position ☐

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0

GEOMETRY QUEST · WORLD 13

Constructions

WORKSHEET



Name _____ Date _____ Stars I give myself: ☆ ☆ ☆

1 Construct an equilateral triangle with sides of 6 cm. Only measure the first side; use your compass for the rest. Leave your arcs showing!

1. Draw AB = 6 cm.
2. Compass on A, open to B. Draw an arc.
3. Same opening, compass on B. Draw an arc.
4. Where the arcs cross is C. Join AC and BC.

2 Can these three sticks make a triangle? Write yes, no, or flat.

3, 4, 5 cm	**2, 2, 5 cm**	**5, 5, 10 cm**	**6, 7, 8 cm**
_____	_____	_____	_____

3 Bisect (cut exactly in half) this 8 cm line with a compass: open it more than halfway, draw arcs above and below from each end, and join the crossings.



Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0

4 Construct a triangle with sides 5 cm, 6 cm and 7 cm. (Draw the 7 cm side, then use arcs of 5 cm and 6 cm from its ends.)

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0

GEOMETRY QUEST · WORLD 14

Congruence

WORKSHEET



Name _____ Date _____ Stars I give myself: ☆ ☆ ☆

1 Two triangles share these facts. Which rule proves they are congruent: SSS, SAS, ASA, AAS, RHS or “not enough”?

all three sides match _____

two sides and the angle between them match _____

two angles and the side between them match _____

a right angle, the hypotenuse and one other side match _____

all three angles match _____

two sides and an angle NOT between them match _____

2 △ABC ≅ △PQR. AB = 5 cm, BC = 7 cm and angle A = 40°. Find PQ, QR and angle P.

PQ = _____ **QR =** _____ **∠P =** _____

3 Draw two triangles that both have angles of 50°, 60° and 70° but are different sizes. What does this show?

4 Why does a garden gate have a diagonal piece of wood across it?

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0

GEOMETRY QUEST · WORLD 15

Pythagoras & √2

WORKSHEET

Name _____ Date _____ Stars I give myself: ☆ ☆ ☆

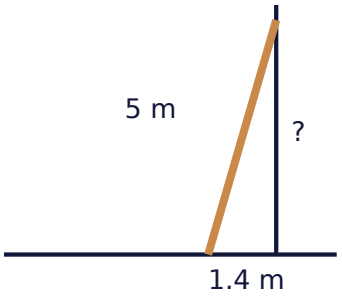
1 Find the hypotenuse (longest side) of each right-angled triangle. $a^2 + b^2 = c^2$.

3 and 4	5 and 12	8 and 15	9 and 12
_____	_____	_____	_____

2 Find the missing shorter side.

hypotenuse 10, side 6	hypotenuse 13, side 5	hypotenuse 25, side 7
_____	_____	_____

3 A 5 m ladder leans on a wall. Its foot is 1.4 m from the wall. How high up the wall does it reach?



_____ m

4 Is a triangle with sides 7, 24 and 25 right-angled? Show your working.

5 Paper investigation: measure a sheet of A4. Divide the long side by the short side. Fold it in half and do the same for the new A5 size.

A4: _____ A5: _____

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0

GEOMETRY QUEST · WORLD 16

Fractals

WORKSHEET



Name _____ Date _____ Stars I give myself: ☆ ☆ ☆

1 Cantor set: each step removes the middle third of every piece. Complete the table.

step	pieces	length left (start = 1)
0	1	1
1	2	2/3
2		
3		
4		

2 Koch snowflake: start with a triangle of side 9 cm. Each step, every edge becomes 4 edges, each a third as long. Complete the table.

step	edges	perimeter
0	3	27 cm
1		
2		
3		

3 Draw step 1 of the Koch curve on this 9 cm line: split it into three 3 cm parts, draw an equilateral triangle on the middle part, then rub out its base.

---

4 Fractal hunt: find three things in nature or at home that repeat their pattern when you zoom in. Draw one of them.

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0